Universal Origins: The Zero-Sum Constraint Matrix Formulation of Cosmogenesis

Infinite-Dimensional Linear Algebra Structure of the Pre-Planck Vacuum and Its Consequences for Big Bang Initial Conditions

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Abstract

We develop a matrix-algebraic framework for cosmogenesis based on a single structural constraint on the universe's density operator. Let $\mathcal{F} = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$ be the Fock space of possible universe states, $\hat{\rho}(t) \in \mathcal{B}(\mathcal{F})$ the global density operator, and $\{\hat{Q}_k\}$ the complete set of Standard Model conserved charges. We impose one axiom: $\mathrm{Tr}\left[\hat{\rho}(t)\hat{Q}_k\right] = 0$ for all k and all times t — a constraint that every row of the observable vector $\mathbf{Q}[\hat{\rho}]$ must equal zero, which we write $\mathbf{Q}[\hat{\rho}] = \mathbf{0}_\infty$ (the infinite zero matrix).

Working strictly within this axiom and the Heisenberg uncertainty principle applied to the Standard Model degree-of-freedom count $g_* = 106.75$, we derive — without additional assumptions — the following results that standard Big Bang cosmology takes as inputs: the radiation equation of state $w = 1/3$, the Hubble rate $H(t_{\mathrm{Pl}}) = 1/(2t_{\mathrm{Pl}})$, exact spatial flatness $\Omega = 1$, and the expansion law $V(t) = (4\pi/3)\ell_{\mathrm{Pl}}^3[1 + (ct/\ell_{\mathrm{Pl}})^2]^{3/2}$. We also present a path-integral saddle-point analysis suggesting that simultaneous emergence of all Planck cells is exponentially preferred over localised fluctuations as a possible mechanism for CMB uniformity, though a complete proof is left as an open problem. Throughout, we distinguish clearly between results that follow from the axiom and those that require additional physics beyond TUO's present scope.

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1 Introduction

The two deepest unanswered questions in cosmology are: *why did the universe begin?* and *why did it begin with those particular initial conditions?* The second question is, in a sense, more tractable. Standard Big Bang cosmology describes the evolution of the universe extraordinarily well from $t \sim t_{\mathrm{Pl}}$ onwards, but its initial conditions — the equation of state $w = 1/3$, the spatial flatness $\Omega = 1$, and the temperature $T \approx T_{\mathrm{Pl}}$ — are inputs rather than outputs [7, 8]. The inflationary scenario addresses some of these by introducing an additional scalar field, but it does not eliminate the question of pre-inflationary initial conditions [4, 5].

The present paper develops the Theory of Universal Origins (TUO), which proposes that the relevant initial conditions follow mathematically from a single structural constraint on the universe's density operator in an infinite-dimensional Fock space. The constraint is expressed as: the trace of the density operator against any conserved charge operator must equal zero at all times. Symbolically, the observable vector built from all conserved charges equals the zero object — the infinite zero matrix. Everything, in a precise operator-algebraic sense, sums to nothing.

The mathematical language we adopt is that of infinite-dimensional linear algebra. The density operator is an element of $\mathcal{B}(\mathcal{F})$; the charge operators are its diagonal entries in an appropriate basis; the constraint is a collection of linear equations on those diagonal entries. This language makes the structure of the constraint — and what it does and does not determine — maximally transparent.

We emphasise what this paper does *not* claim:

- We do not claim to explain the number of fermion generations.
- We do not claim to derive the baryon asymmetry η quantitatively.
- We do not claim to explain the CMB power spectrum.
- We do not claim to replace general relativity; a companion framework (Quantum Gravity Dynamics) that recovers GR from the flat background is required and is stated as a separate project.
- We do not derive dark matter or dark energy from these axioms.
- Where a result requires physics beyond the two axioms, we explicitly label it as a *possible mechanism* or *open problem*.

2 Mathematical Foundations: The Zero Matrix and Fock Space

2.1 The zero object

The central object of TUO is the infinite zero matrix:

$$\mathbf{0}_\infty = \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \in \mathcal{B}(\mathcal{F}). \tag{1}$$

The claim of TUO is that the observable content of the universe, when correctly assembled, equals $\mathbf{0}_\infty$ at every instant. This is not a dynamical statement; it is an existence filter.

2.2 Fock space structure

Definition 2.1 (Universe Fock space)**.** Let $\mathcal{F} = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$ be the bosonic-fermionic Fock space over the single-particle Hilbert space $\mathcal{H}_1$ of Standard Model fields. States $|\Psi\rangle \in \mathcal{F}$ are superpositions of n-particle sectors:

$$|\Psi\rangle = \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \end{pmatrix}, \quad \sum_{n=0}^{\infty} |c_n|^2 = 1. \tag{2}$$

2.3 Density operator

The universe is described by a density operator $\hat{\rho}(t) \in \mathcal{B}(\mathcal{F})$ satisfying

$$\hat{\rho} = \hat{\rho}^\dagger, \quad \mathrm{Tr}[\hat{\rho}] = 1, \quad \langle\psi|\hat{\rho}|\psi\rangle \geq 0 \; \forall \, |\psi\rangle. \tag{3}$$

In a basis $\{|n\rangle\}_{n=0}^{\infty}$ of $\mathcal{F}$:

$$\hat{\rho} = \sum_{m,n=0}^{\infty} \rho_{mn} |m\rangle\langle n| = \begin{pmatrix} \rho_{00} & \rho_{01} & \rho_{02} & \cdots \\ \rho_{10} & \rho_{11} & \rho_{12} & \cdots \\ \rho_{20} & \rho_{21} & \rho_{22} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4}$$

2.4 Charge operators

Each conserved charge corresponds to a Hermitian operator $\hat{Q}_k \in \mathcal{B}(\mathcal{F})$ with eigenvalues $\{q_{kn}\}_{n\geq 0}$. In the eigenbasis of $\hat{Q}_k$:

$$\hat{Q}_k = \sum_{n=0}^{\infty} q_{kn} |n\rangle\langle n| = \begin{pmatrix} q_{k0} & 0 & 0 & \cdots \\ 0 & q_{k1} & 0 & \cdots \\ 0 & 0 & q_{k2} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{5}$$

The expectation value is the trace:

$$\langle \hat{Q}_k \rangle = \mathrm{Tr}\Big[\hat{\rho}\, \hat{Q}_k\Big] = \sum_{n=0}^{\infty} \rho_{nn}\, q_{kn}. \tag{6}$$

This is an *infinite dot product* of the occupation-probability vector $\mathbf{p} = (\rho_{00}, \rho_{11}, \rho_{22}, \ldots)$ against the eigenvalue vector $\mathbf{q}_k = (q_{k0}, q_{k1}, q_{k2}, \ldots)$.

3 The Two Axioms

Axiom 1 (Flat Background)**.** The pre-emergence arena is (3+1)-dimensional Minkowski spacetime: $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$. No curvature, no preferred time, no pre-existing matter.

Remark 3.1. Axiom 1 is foundational and self-standing. It is not derived from general relativity, string theory, or any other framework. Why four dimensions? Why Minkowski rather than another flat geometry? These questions are not answered here; the axiom names the background on which the theory is defined, and treats that choice as primitive.

Axiom 2 (Global Zero-Sum Constraint)**.** For all conserved charges $\hat{Q}_k$ and all times t:

$$\boxed{\mathrm{Tr}\big[\hat{\rho}(t)\, \hat{Q}_k\big] = 0 \quad \forall\, k,\ \forall\, t.} \tag{7}$$

Equivalently, the observable charge vector

$$\mathbf{Q}[\hat{\rho}] = \begin{pmatrix} \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_E\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{P^x}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{P^y}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{P^z}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{L^x}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{L^y}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{L^z}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{\mathrm{em}}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{B-L}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{C_r}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{C_g}\right] \\ \mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_{C_b}\right] \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \mathbf{0}_{12\times 1}. \tag{8}$$

Remark 3.2. Energy (the Hamiltonian) is one conserved charge, associated with time-translation symmetry by Noether's theorem. Axiom 2 is therefore strictly stronger than requiring zero total energy alone: it also sets net momentum, angular momentum, and every gauge charge to zero. The pre-emergence state is a quantum nothing in every sense that Noether's theorem can assign a conserved number to.

Imposing zero energy alone would leave the possibility of a state with, for example, nonzero angular momentum — which would carry a preferred axis into the emerged universe. The full zero-sum axiom eliminates all such remnants simultaneously.

Remark 3.3. Equation (7) is an infinite linear constraint on $\hat{\rho}$: substituting (6), it reads $\sum_{n=0}^{\infty} \rho_{nn}\, q_{kn} = 0$ for each k. This defines a hyperplane $\mathcal{Z}_k \subset \mathcal{B}(\mathcal{F})$, and the set of physically allowed density operators is $\mathcal{Z} = \bigcap_k \mathcal{Z}_k \cap \{\text{density operators}\}$.

Remark 3.4. Axiom 1 and Axiom 2 are the *only* postulates of TUO. Every other result in this paper is derived from these two axioms plus Standard Model particle content. Where we invoke additional physical input, it is explicitly indicated.

4 Vacuum State and the Zero-Sum Filter

4.1 The vacuum

Definition 4.1 (Vacuum state)**.** The ground state prior to any excitation:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \qquad \hat{\rho}_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{9}$$

Lemma 4.2. *The vacuum state satisfies Axiom 2.*

Proof. $\mathrm{Tr}\left[\hat{\rho}_0 \hat{Q}_k\right] = q_{k0} = 0$ (vacuum carries no charge). □

4.2 Single-particle prohibition

Theorem 4.3. *A pure single-particle state $|1; q\rangle$ with charge eigenvalue $q \neq 0$ violates Axiom 2.*

Proof. Let $|1; q\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}$, $\hat{\rho}_1 = |1; q\rangle\langle 1; q|$, $\hat{Q} = \mathrm{diag}(0, q, 0, \ldots)$. Then $\mathrm{Tr}\left[\hat{\rho}_1 \hat{Q}\right] = q \neq 0$. □

4.3 Pair configurations

Theorem 4.4. *A superposition of states with opposite charge eigenvalues satisfies Axiom 2 for that charge.*

Proof. Let $|+\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}$, $|-\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \end{pmatrix}$, $|\psi\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$. Then $\hat{\rho}_{\text{pair}} = |\psi\rangle\langle\psi| = \frac{1}{2}\begin{pmatrix} 0 & 0 & \cdots & \\ 0 & 1 & 1 & \cdots \\ 0 & 1 & 1 & \cdots \\ \vdots & & & \end{pmatrix}$ and $\hat{Q} = \mathrm{diag}(0, +q, -q, \ldots)$, giving $\mathrm{Tr}\left[\hat{\rho}_{\text{pair}} \hat{Q}\right] = \frac{1}{2}(q - q) = 0$. □

Remark 4.5. The proof works for any superposition of states whose charge eigenvalues sum to zero. The zero-sum constraint acts as a *linear filter*: it admits any state in the kernel of the charge functional, and rejects all others.

5 The Complete Charge Equation

5.1 Hamiltonian decomposition

The total Hamiltonian splits as $\hat{H} = \hat{H}_{\text{matter}} + \hat{H}_{\text{field}}$, where in the energy eigenbasis:

$$\hat{H}_{\text{matter}} = \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 0 & E_1 & 0 & \cdots \\ 0 & 0 & E_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad E_n > 0. \tag{10}$$

Theorem 5.1 (Energy balance)**.** *Axiom 2 applied to $\hat{H}$ requires*

$$\operatorname{Tr}\Big[\hat{\rho}\,\hat{H}_{\text{matter}}\Big] + \operatorname{Tr}\Big[\hat{\rho}\,\hat{H}_{\text{field}}\Big] = 0, \tag{11}$$

i.e., $E_{\text{field}} = -E_{\text{matter}}$ exactly.

Proof. $\operatorname{Tr}\Big[\hat{\rho}\,\hat{H}\Big] = 0 = \operatorname{Tr}\Big[\hat{\rho}(\hat{H}_{\text{m}} + \hat{H}_{\text{f}})\Big] = E_{\text{m}} + E_{\text{f}}.$ □

5.2 The complete zero equation

All 12 conserved-charge observables form a column vector. Axiom 2 states that this vector equals the zero column:

$$\underbrace{\begin{pmatrix} \operatorname{Tr}\Big[\hat{\rho}\,\hat{Q}_E\Big] \\ \operatorname{Tr}\Big[\hat{\rho}\,\hat{Q}_{P^\mu}\Big] \\ \operatorname{Tr}\Big[\hat{\rho}\,\hat{Q}_J\Big] \\ \operatorname{Tr}\Big[\hat{\rho}\,\hat{Q}_{\text{em}}\Big] \\ \operatorname{Tr}\Big[\hat{\rho}\,\hat{Q}_{B-L}\Big] \\ \operatorname{Tr}\Big[\hat{\rho}\,\hat{Q}_{\mathbf{C}}\Big] \end{pmatrix}}_{\mathbf{Q}[\hat{\rho}] \in \mathbb{R}^{12}} = \begin{pmatrix} 0 \\ \mathbf{0}_4 \\ \mathbf{0}_3 \\ 0 \\ 0 \\ \mathbf{0}_3 \end{pmatrix} = \mathbf{0}_{12\times 1}. \tag{12}$$

This is the *central equation of TUO*: the universe's full observable content equals the infinite zero.

Remark 5.2. The zero on the right-hand side is not an approximation. It is exact and holds at all times by Axiom 2. The field contribution is whatever it must be to maintain the balance: for each matter configuration with charges $\mathbf{Q}_{\text{m}}$, the field sector must carry $\mathbf{Q}_{\text{f}} = -\mathbf{Q}_{\text{m}}$.

5.3 Time evolution preserves zero-sum

Theorem 5.3. *Unitary evolution $\hat{\rho}(t) = \hat{U}(t)\hat{\rho}(0)\hat{U}^\dagger(t)$ preserves Axiom 2 if and only if $[\hat{H}, \hat{Q}_k] = 0$ for all k.*

Proof. $\frac{d}{dt}\operatorname{Tr}\left[\hat{\rho}\hat{Q}_k\right] = \operatorname{Tr}\left[\frac{\partial\hat{\rho}}{\partial t}\hat{Q}_k\right] = -\frac{i}{\hbar}\operatorname{Tr}\left[[\hat{H}, \hat{\rho}]\hat{Q}_k\right] = -\frac{i}{\hbar}\operatorname{Tr}\left[\hat{\rho}[\hat{Q}_k, \hat{H}]\right]$. This vanishes if and only if $[\hat{H}, \hat{Q}_k] = 0$, i.e., each $\hat{Q}_k$ is conserved. □

6 The Selection Problem: Why Generic Fluctuations Cannot Become a Universe

Axiom 2 permits a vast class of zero-sum fluctuations from the vacuum. Theorems 4.3 and 4.4 show that the constraint acts as a filter: single charged particles are excluded, but any superposition whose charges cancel is admitted. The vacuum can fluctuate. The question is which fluctuations can survive long enough, and carry enough energy, to become a universe. Two explicit barriers prevent generic fluctuations from doing so.

6.1 Barrier I: The energy gap

A typical virtual pair — an electron and positron — carries energy

$$E_{e^+e^-} = 2m_ec^2 \approx 1.64 \times 10^{-13}\ \text{J}. \tag{13}$$

The energy required to seed a Planck-scale cell capable of expanding into a universe is (Section 9):

$$E_{\text{cell}} = g_* \cdot \frac{E_{\text{Pl}}}{2} \approx 1.04 \times 10^{11}\ \text{J}. \tag{14}$$

The ratio is

$$\frac{E_{e^+e^-}}{E_{\text{cell}}} \approx 1.6 \times 10^{-24}. \tag{15}$$

A generic virtual pair carries twenty-four orders of magnitude too little energy. No accumulation mechanism exists within a single Planck-time window to bridge this gap from below.

6.2 Barrier II: The annihilation timescale

The Heisenberg time-energy relation constrains the lifetime of any virtual configuration:

$$\Delta t \sim \frac{\hbar}{\Delta E}. \tag{16}$$

At Planck energies, $\Delta t \sim t_{\text{Pl}} \approx 5.4 \times 10^{-44}$ s. Within this window, a generic fluctuation appears and annihilates before any complex structure can form. Even if an arbitrary

collection of particle-antiparticle pairs were to emerge, each pair presents its own annihilation channel — $e^+e^- \to \gamma\gamma$, $q\bar{q} \to gg$, and so on — and the configuration collapses back to the vacuum on the Planck timescale.

6.3 The resolution: the maximum fluctuation

The two barriers are simultaneously and uniquely overcome by the most extreme fluctuation consistent with Axiom 2: *all Standard Model degrees of freedom emerging at the same point in space at the same moment in time.* Not sequentially. Not spread out. All of them, simultaneously, co-located at a single Planck-scale volume.

Concretely, this is a configuration in which all $g_* = 106.75$ effective SM degrees of freedom are excited to their Heisenberg minimum energy $E_{\rm Pl}/2$, co-located in one Planck cell, simultaneously.

Theorem 6.1 (Maximum fluctuation uniquely overcomes both barriers)**.** *The configuration consisting of all SM matter — quarks and leptons across three generations, in all colours and flavours — simultaneously co-located at one Planck cell, with no antiparticles:*

(i) satisfies Axiom 2, and

(ii) carries energy $E_{\rm cell} = g_* \cdot E_{\rm Pl}/2$, and

(iii) has no annihilation channel and is therefore stable.

Proof. (i) Zero-sum satisfied. The SM fermion content satisfies $B - L = 0$ and $Q = 0$ per generation (Theorem 8.1 below); the total vanishes for all three generations. Gravitational energy $E_{\rm grav} = -E_{\rm cell}$ by Theorem 5.1. All twelve charges in Eq. (8) vanish.

(ii) Energy. Each of the g_* modes carries $E_{\rm Pl}/2$ from the Heisenberg bound. Total: $E_{\rm cell} = g_* \cdot E_{\rm Pl}/2$ (Theorem 9.1 below).

(iii) Stability. The configuration contains only matter: antiparticle occupation numbers $n_{\bar{f}} = 0$ for all $\bar{f}$. By Theorem 8.3 below, no SM process can open an annihilation channel. The configuration is kinematically stable. □

Remark 6.2. The resolution is not a timescale argument. It is a *kinematic* one. Annihilation requires a particle and its antiparticle. The maximum fluctuation contains no antiparticles. There is no annihilation, not because the timescale is too short, but because the required reactant is absent by construction. This is what the No-Annihilation Theorem (Theorem 8.3) will make precise.

Remark 6.3. This is the genuine creative core of TUO. The zero-sum vacuum is not inert: it fluctuates. Generically its fluctuations are too small and too short-lived — the two barriers above are absolute. The unique exception is the configuration that carries all

SM content, simultaneously, co-located: it is energetically sufficient and kinematically stable. It is the only zero-sum fluctuation the vacuum can produce that is capable of becoming a universe. The number $g_* = 106.75$ enters not as a free parameter, but as the total count of SM degrees of freedom that must all emerge together for either barrier to be overcome.

7 Configuration Space and Frozen Constants

The maximum fluctuation selects *a* configuration, not necessarily *the only* configuration. This section characterises the full space of zero-sum configurations that can produce stable universes, and identifies which aspects of physical law are fixed by the axioms versus fixed by the particular configuration that emerged.

7.1 The space of stable configurations

A configuration is a choice of field content (determining g_*), charge assignments (determining whether Axiom 2 is satisfiable with no antiparticles), and a mass spectrum. The zero-sum filter admits a matter-only configuration if and only if it satisfies the anomaly-cancellation conditions: $Q = 0$ and $B - L = 0$ per generation-like block. These are identical to the SU(5) grand-unification consistency conditions, which the Standard Model satisfies exactly (Theorem 8.1).

Other configurations with $N_{\text{gen}} \neq 3$ also satisfy these conditions, since the anomaly-cancellation identity holds *per generation* independently. The corresponding g_* values and cell energies are shown in Table 1.

Table 1: Stable zero-sum configurations for varying number of generations. Each configuration satisfies $Q = 0$, $B - L = 0$, passes the no-annihilation test, and could in principle become a universe with the given cell energy. N_ν is measured at LEP [3]: only $N_{\text{gen}} = 3$ is consistent with observation.

N_{gen}	g_*	$E_{\text{cell}}/E_{\text{Pl}}$	E_{cell} (J)
1	42.00	21.00	4.11×10^{10}
2	56.00	28.00	5.48×10^{10}
3	**70.00**	**35.00**	$\mathbf{6.85 \times 10^{10}}$
4	84.00	42.00	8.22×10^{10}
5	98.00	49.00	9.58×10^{10}

Remark 7.1. Note: the g_* values in Table 1 count only fermion generations; the bosonic sector ($N_B = 28$ for the SM gauge and Higgs bosons) may also differ across configurations. The value $g_* = 106.75$ in Table 4 includes the full SM boson content.

Remark 7.2. Each row of Table 1 describes a logically consistent configuration that satisfies both axioms. They correspond to different possible universes. The observation $N_\nu = 2.9840 \pm 0.0082$ (LEP) selects $N_{\text{gen}} = 3$ from among these possibilities. TUO does not derive $N_{\text{gen}} = 3$; this is an open problem (Section 17).

7.2 What freezes at emergence: a classification

When a configuration emerges, three classes of quantities are fixed.

Class I — Axiom-fixed constants. The speed of light c is fixed by Axiom 1: Minkowski spacetime has a unique light-cone structure, and c is its invariant speed. The reduced Planck constant $\hbar$ is required for the Heisenberg bound in Axiom 2. Newton's constant G is required by Theorem 5.1: the gravitational energy $E_{\text{grav}} = -GM_{\text{Pl}}^2/(2\ell_{\text{Pl}})$ must exactly cancel $E_{\text{matter}} = E_{\text{Pl}}/2$, which forces $GM_{\text{Pl}}^2 = \hbar c$ — an identity that defines G in terms of $\hbar$ and c. These three constants are *not* configuration-dependent; they are consequences of the axioms themselves.

Class II — Configuration-fixed dimensionless ratios. The fine-structure constant $\alpha \approx 1/137$, the strong coupling α_s, and the mass ratios m_f/M_{Pl} are fixed when a particular configuration is selected. A universe with $N_{\text{gen}} = 4$ would, in general, have different values of these ratios. These are the parameters of the effective Lagrangian and they vary across configurations.

Class III — Emergent effective constants. The cosmological constant Λ, the Higgs vacuum expectation value, and the QCD scale Λ_{QCD} are dynamically determined within a configuration. They are consequences of Class II, not independent choices.

Remark 7.3. A common intuition is that the speed of light could be different in another universe. TUO refines this: c as a *speed limit* cannot vary because it is fixed by Axiom 1. What can vary is the ratio $m_e c^2/E_{\text{Pl}}$ — the electron is lighter or heavier relative to the Planck scale — which would alter all of chemistry, not the geometry of spacetime.

8 Standard Model Content and the $B-L$ Constraint

8.1 Why $B - L$, not B alone

TUO constrains $B - L = 0$ rather than $B = 0$ separately. The constraint $B = 0$ would require equal numbers of quarks and antiquarks, forbidding matter-only universes. The constraint $B - L = 0$ permits matter-only configurations in which $B = L \neq 0$, provided their difference vanishes.

Theorem 8.1. *The Standard Model three-generation fermion content satisfies $Q = 0$ and $B - L = 0$ per generation as an algebraic identity.*

Proof. In generation g, denote the quark doublet (u_g, d_g) in three colour copies and the lepton doublet (ν_g, e_g).

Baryon minus lepton number:
$B_g = 2\,\text{flavours} \times 3\,\text{colours} \times \frac{1}{3} = 2$; $L_g = 2\,\text{leptons} \times 1 = 2$; therefore $B_g - L_g = 0$.

Electric charge:

$$Q_g = 3\left(\tfrac{2}{3}\right) + 3\left(-\tfrac{1}{3}\right) + (-1) + 0 = 2 - 1 - 1 = 0.$$

Both identities hold for $g = 1, 2, 3$ by the same counting. □

Remark 8.2. This result is the anomaly-cancellation condition of SU(5) grand unification. TUO does not derive the SM content; rather, the SM's structure satisfies TUO's constraint. Whether TUO can derive the SM content from first principles remains an open question (Section 17).

8.2 Matter-only stability

Theorem 8.3 (No-Annihilation)**.** *A configuration satisfying $Q = 0$, $B - L = 0$, and containing no antiparticles has no SM annihilation channel.*

Proof. Every SM annihilation reaction requires a particle-antiparticle pair: $e^-e^+ \to \gamma\gamma$, $q\bar{q} \to gg$, etc. In a configuration with zero antiparticle occupation ($n_{\bar{f}} = 0$ for all $\bar{f}$), no such pair exists. No SM process can produce $\bar{f}$ from an initial state with $n_{\bar{f}} = 0$ without violating lepton or baryon number conservation. □

Remark 8.4. The argument is purely kinematic: the absence of antiparticles removes the kinematic possibility of annihilation. No claim is made about interaction timescales or thermodynamic equilibrium.

9 Energy from Heisenberg Uncertainty

9.1 The energy matrix at $t = t_{\text{Pl}}$

The Heisenberg time-energy uncertainty relation gives the minimum energy per mode at timescale $\Delta t = t_{\text{Pl}}$:

$$\Delta E\,\Delta t \geq \frac{\hbar}{2} \implies \Delta E_{\min} = \frac{\hbar}{2t_{\text{Pl}}} = \frac{E_{\text{Pl}}}{2}. \tag{17}$$

At $T = T_{\text{Pl}}$, all SM fields are relativistic. The number of relativistic degrees of freedom is

$$g_* = N_B + \tfrac{7}{8}N_F = 28 + \tfrac{7}{8}(90) = 106.75, \tag{18}$$

where $N_B = 28$ (gauge and Higgs bosons) and $N_F = 90$ (fermionic polarisations and colour states).

The Hamiltonian for the emergence configuration, restricted to the first g_* excitation modes at their minimum Heisenberg energies, takes the form:

$$\hat{H}_{\text{emergence}} = \begin{pmatrix} 0 & & & & \\ & \frac{E_{\text{Pl}}}{2} & & & \\ & & \frac{E_{\text{Pl}}}{2} & & \\ & & & \ddots & \\ & & & & \frac{E_{\text{Pl}}}{2} \end{pmatrix}_{g_* \times g_*} \oplus \mathbf{0}_{\infty - g_*}, \tag{19}$$

with g_* copies of $E_{\text{Pl}}/2$ on the diagonal.

Theorem 9.1 (Energy per Planck cell)**.** *The total matter energy per Planck cell is*

$$\boxed{E_{\text{cell}} = g_* \cdot \frac{E_{\text{Pl}}}{2} = 53.375\, E_{\text{Pl}}.} \tag{20}$$

By Theorem 5.1, the gravitational field energy satisfies $E_{\text{grav}} = -E_{\text{cell}}$ exactly. The total energy of the universe is zero.

Proof. Each of the g_* modes carries minimum energy $E_{\text{Pl}}/2$ from Eq. (17). The sum is immediate. $E_{\text{grav}} = -E_{\text{cell}}$ follows from Theorem 5.1 applied to $\hat{H}_{\text{field}} = \hat{H}_{\text{grav}}$. □

9.2 Comparison with standard cosmology

The energy density of a thermal plasma at $T = T_{\text{Pl}}$ in volume ℓ_{Pl}^3, from the Standard Model equation of state, is

$$E_{\text{std}} = \frac{\pi^2}{30}\, g_* \, \frac{(k_B T_{\text{Pl}})^4}{(\hbar c)^3}\, \ell_{\text{Pl}}^3 = 6.87 \times 10^{10}\ \text{J}. \tag{21}$$

The TUO value is $E_{\text{cell}} = 53.375\, E_{\text{Pl}} = 1.044 \times 10^{11}$ J. Their ratio is

$$\frac{E_{\text{cell}}}{E_{\text{std}}} = \frac{15}{\pi^2} \approx 1.52. \tag{22}$$

This factor is a pure dimensionless number independent of all SM parameters. It is the ratio of the Heisenberg minimum energy to the thermal average energy per mode.

10 Simultaneous Emergence: A Possible Mechanism

Remark 10.1. The analysis in this section is a *possible mechanism*, not a proven theorem. The saddle-point argument below suggests that simultaneous emergence of all Planck

cells is exponentially preferred over localised fluctuations under the constrained measure, but a rigorous proof — with controlled error bounds on the saddle-point approximation and a connection to the observed $\Delta T/T \sim 10^{-5}$ — is left as Open Question 17.2.

10.1 Setup

Partition the Minkowski background (Axiom 1) into a lattice of Planck cells indexed by $i = 1, \ldots, N$. The number of cells in the observable universe's comoving volume at $t = t_{\mathrm{Pl}}$ is

$$N = \left(\frac{R_0 \sqrt{t_{\mathrm{Pl}}/t_0}}{\ell_{\mathrm{Pl}}} \right)^3 \approx 8.8 \times 10^{92}. \tag{23}$$

The constrained Euclidean path integral is

$$Z = \int \mathcal{D}\phi \, e^{-S_E[\phi]} \prod_k \delta\big(Q_k[\phi]\big), \tag{24}$$

where the delta functions enforce Axiom 2 globally.

10.2 Saddle-point analysis

Writing $\delta(Q_k) = \int \frac{d\lambda_k}{2\pi} e^{i\lambda_k Q_k}$:

$$Z = \int \mathcal{D}\phi \, \mathcal{D}\lambda \, \exp\left(-S_E[\phi] + i \sum_k \lambda_k Q_k[\phi] \right). \tag{25}$$

At the saddle point, single-cell fluctuations against $N-1$ vacuum cells incur an action penalty $\Delta S \sim N$, while simultaneous N-cell emergence satisfies the global constraint with $\Delta S = 0$. The ratio of partition-function contributions is formally

$$\frac{Z_{\mathrm{simultaneous}}}{Z_{\mathrm{single}}} \sim e^N = e^{8.8 \times 10^{92}}. \tag{26}$$

This suggests simultaneous emergence is the overwhelmingly dominant configuration, offering a possible explanation for the observed uniformity of the CMB. A full proof, controlling the saddle-point approximation and connecting the amplitude to the observed $\Delta T/T$, is left as Open Question 17.2.

11 Expansion Law from Quantum Wavepacket Spreading

11.1 Wavepacket initial condition

At $t = t_{\mathrm{Pl}}$, the emergent configuration is localised to one Planck cell:

$$\psi(\mathbf{x}, 0) = (2\pi\ell_{\mathrm{Pl}}^2)^{-3/4} \exp\left(-\frac{x^2}{4\ell_{\mathrm{Pl}}^2}\right). \tag{27}$$

Position-momentum uncertainty with $\Delta x = \ell_{\mathrm{Pl}}$:

$$\Delta p \geq \frac{\hbar}{2\ell_{\mathrm{Pl}}} = \frac{M_{\mathrm{Pl}}c}{2}. \tag{28}$$

Relativistic velocity from $E = E_{\mathrm{Pl}}/2$, $p = M_{\mathrm{Pl}}c/2$:

$$v_0 = \frac{pc^2}{E} = \frac{(M_{\mathrm{Pl}}c/2)c^2}{E_{\mathrm{Pl}}/2} = c. \tag{29}$$

This immediately gives $w = P_{\mathrm{kinetic}}/(uc^2) = 1/3$ (radiation), which TUO derives rather than assumes.

11.2 Derived expansion law

Theorem 11.1 (Expansion law)**.** *For massless particles ($v = c$), the wavepacket width evolves as*

$$\sigma(t) = \ell_{\mathrm{Pl}}\sqrt{1 + \left(\frac{ct}{\ell_{\mathrm{Pl}}}\right)^2}, \tag{30}$$

giving the volume

$$V(t) = \frac{4\pi}{3}\,\sigma(t)^3 = \frac{4\pi}{3}\,\ell_{\mathrm{Pl}}^3\left[1 + \left(\frac{ct}{\ell_{\mathrm{Pl}}}\right)^2\right]^{3/2}. \tag{31}$$

For $ct \gg \ell_{\mathrm{Pl}}$:

$$V(t) \approx \frac{4\pi}{3}(ct)^3\left[1 + \frac{\Delta V}{V}\right], \quad \frac{\Delta V}{V} = 3\left(\frac{\ell_{\mathrm{Pl}}}{ct}\right)^2. \tag{32}$$

Proposition 11.2 (Sub-luminal bound)**.** *$v(t) = d\sigma/dt < c$ for all $t > 0$.*

Proof. $v(t) = c(ct/\ell_{\mathrm{Pl}})/\sqrt{1 + (ct/\ell_{\mathrm{Pl}})^2} < c$. □

Remark 11.3. The expansion speed in TUO is sub-luminal at all times. Standard inflation requires super-luminal expansion of the scale factor. This is a distinguishing feature of TUO.

11.3 Hubble parameter

$$H_{\mathrm{TUO}}(t) = \frac{1}{\sigma}\frac{d\sigma}{dt} = \frac{c^2 t}{\ell_{\mathrm{Pl}}^2 + c^2 t^2}. \tag{33}$$

At $t = t_{\mathrm{Pl}}$, using $\ell_{\mathrm{Pl}} = ct_{\mathrm{Pl}}$:

$$H_{\mathrm{TUO}}(t_{\mathrm{Pl}}) = \frac{c^2 t_{\mathrm{Pl}}}{2c^2 t_{\mathrm{Pl}}^2} = \frac{1}{2t_{\mathrm{Pl}}} = H_{\mathrm{FRW}}(t_{\mathrm{Pl}}). \tag{34}$$

The Hubble parameter is continuous across the TUO–FRW junction.

12 Junction Conditions at $t = t_{\mathrm{Pl}}$

Theorem 12.1 (Smooth junction)**.** *The TUO solution at $t = t_{\mathrm{Pl}}^-$ matches the FRW radiation-dominated solution at $t = t_{\mathrm{Pl}}^+$ with no discontinuity in H, w, or k.*

Proof. Three conditions must be checked:

(i) *Hubble parameter*: verified in Eq. (34). ✓

(ii) *Equation of state*: TUO gives $w = 1/3$ from Eq. (29); FRW takes $w = 1/3$ as input. TUO derives what FRW assumes. ✓

(iii) *Spatial curvature*: Axiom 1 requires $k = 0$ exactly; FRW allows $k = 0$; Planck 2018 measures $\Omega = 1.0007 \pm 0.0037$ [1]. ✓

No surface energy term is required. □

Remark 12.2. The energy density comparison (junction condition IV) gives a mismatch by a factor $15/\pi^2 \approx 1.52$. This does not break the smooth junction for H, w, k; it reflects the difference between the TUO Heisenberg minimum-energy state and the FRW thermal-equilibrium state. The physical interpretation of this factor is left as an open problem.

13 Thermal History

Theorem 13.1 (Pre-emergence temperature)**.** *The temperature T_{TUO} of a radiation bath with g_* degrees of freedom having the same energy density as the TUO pre-emergence state is*

$$T_{\mathrm{TUO}} = \left(\frac{15}{\pi^2}\right)^{1/4} T_{\mathrm{Pl}} \approx 1.11\, T_{\mathrm{Pl}}, \tag{35}$$

independent of g_*.

Proof. Set $\rho_{\mathrm{TUO}} = g_* \cdot E_{\mathrm{Pl}}/(2\ell_{\mathrm{Pl}}^3)$ equal to the Stefan-Boltzmann density $\rho_{\mathrm{SB}} = (\pi^2/30) g_* (k_B T)^4/(\hbar c)^3$. The factor g_* cancels on both sides, yielding $(k_B T)^4 = (15/\pi^2)(k_B T_{\mathrm{Pl}})^4$. □

Remark 13.2. The g_*-independence is non-trivial: the particle content determines the total energy but not the temperature, because both the Heisenberg energy and the Stefan-Boltzmann density scale identically with g_*. The temperature after the junction follows the standard radiation-dominated law $T(t) = T_{\mathrm{TUO}}\sqrt{t_{\mathrm{Pl}}/t}$.

14 Equality in Fundamental Equations as a Consequence of the Zero-Sum Axiom

A striking feature of fundamental physics is that its governing equations are *equalities*: $G_{\mu\nu} = \kappa T_{\mu\nu}$, not $G_{\mu\nu} \geq \kappa T_{\mu\nu}$. This is not obvious. Why should geometry exactly balance matter-energy, rather than exceed or fall short of it? TUO provides a structural answer.

14.1 Why equalities, not inequalities

Axiom 2 states $\mathrm{Tr}\left[\hat{\rho}\,\hat{Q}_k\right] = 0$ *exactly* for every conserved charge. This propagates: for any local density $\mathcal{Q}_k(x)$ associated with $\hat{Q}_k$,

$$\int \mathcal{Q}_k(x)\, d^3x = 0 \quad \forall\, t. \tag{36}$$

If at any spacetime point x_0 the local charge density were positive, Axiom 2 requires it to be exactly compensated elsewhere. Localising this balance at every point requires

$$\mathcal{Q}_k^{\mathrm{matter}}(x) + \mathcal{Q}_k^{\mathrm{field}}(x) = 0 \quad \forall\, x. \tag{37}$$

This is an equality because any surplus or deficit at a single point violates the axiom. The sign = is not empirical; it is forced.

14.2 The Einstein field equations

Applied to the energy-momentum charge, Theorem 5.1 gives $E_{\mathrm{matter}} + E_{\mathrm{grav}} = 0$ globally. Localising yields $T_{\mu\nu}^{\mathrm{matter}}(x) + T_{\mu\nu}^{\mathrm{grav}}(x) = 0$ pointwise. By the Lovelock theorem [9], the unique rank-2 symmetric divergence-free tensor constructible from the metric alone is $aG_{\mu\nu} + bg_{\mu\nu}$; matching the Newtonian limit fixes $a = 8\pi G/c^4$. One therefore has:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}. \tag{38}$$

The left side is the gravitational charge density; the right side is the matter charge density. Their sum is zero: this is the Einstein field equation, and its equality sign is the local expression of Axiom 2. TUO does not *derive* the Lovelock theorem; rather, given that theorem, the EFE structure is *forced* by the axiom.

14.3 Maxwell's equations

For the electromagnetic charge Q_{em}, the local zero-sum reads $\partial_\mu F^{\mu\nu}(x) + J^\nu(x)/\varepsilon_0 = 0$, i.e.,

$$\partial_\mu F^{\mu\nu} = \frac{J^\nu}{\varepsilon_0}. \tag{39}$$

The field exactly compensates its source. An inequality would mean the electromagnetic field carries more (or less) charge than the source creates, violating Axiom 2.

14.4 The Schrödinger and Dirac equations

For probability conservation ($Q_{\mathrm{prob}} = \int |\psi|^2 d^3x = 1$), Axiom 2 requires this integral to be constant: $d/dt \int |\psi|^2 d^3x = 0$. The unique evolution that achieves this while being linear and first-order in time is $i\hbar\,\partial_t\psi = \hat{H}\psi$. Similarly, the Dirac equation $(i\gamma^\mu\partial_\mu - m)\psi = 0$ enforces zero-sum between positive and negative spinor components.

Remark 14.1. The structure of these observations is the same in each case: Axiom 2 says a conserved charge totals to zero; the corresponding dynamical equation is the local expression of that balance. The equality sign in every fundamental equation is the local expression of $\mathbf{Q}[\hat{\rho}] = \mathbf{0}_\infty$.

14.5 Exotic phenomena permitted by the zero-sum axiom

The axiom constrains but does not over-constrain. Table 2 lists phenomena that satisfy Axiom 2 and are therefore admitted — some observed, some conjectural.

15 Statistical Analysis of the Energy Result

The energy per Planck cell derived in Theorem 9.1,

$$E_{\mathrm{cell}} = \frac{g_*}{2}\,E_{\mathrm{Pl}} = 53.375\,E_{\mathrm{Pl}}, \tag{40}$$

follows from two axioms and the Heisenberg bound, with no free parameters beyond the SM particle count $g_* = 106.75$. This section analyses how extraordinary this result is.

Table 2: Phenomena permitted by the zero-sum axiom. Status: ✓ = observed; ∼ = conjectural direction consistent with ZSC.

Phenomenon	Zero-sum mechanism	Status
Virtual particles	Particle-antiparticle pair: $Q_{\text{pair}} = 0$. Heisenberg limits duration, not existence.	✓
Quantum entanglement	Singlet state enforces $\langle S_z \rangle_A + \langle S_z \rangle_B = 0$ at all separations (Section 13).	✓
CPT symmetry	Every process that creates particles has a CPT-conjugate creating antiparticles; global $B - L = 0$ is maintained.	✓
Black hole information	Information (von Neumann entropy) is a conserved charge; ZSC requires $S_{\text{matter}} + S_{\text{grav}} = \text{const}$: information transfers to the gravitational sector, not destroyed.	∼
Dark sector	Any hidden sector with internal $Q_{\text{dark}} = 0$, $B_{\text{dark}} - L_{\text{dark}} = 0$ satisfies ZSC. Dark matter as a parallel zero-sum sector is admitted.	∼
Parallel configurations	Other N_{gen} configurations (Table 1) satisfy ZSC and could in principle produce stable universes with different dimensionless ratios.	∼

15.1 The universal factor $15/\pi^2$

Setting the TUO energy density equal to the Stefan-Boltzmann density at the same temperature, the factor g_* cancels (Theorem 13.1), leaving:

$$\frac{E_{\text{cell}}}{E_{\text{thermal}}} = \frac{g_* \cdot E_{\text{Pl}}/2}{(\pi^2/30)\, g_* \, (k_B T_{\text{Pl}})^4/(\hbar c)^3 \cdot \ell_{\text{Pl}}^3} = \frac{15}{\pi^2} \approx 1.5198. \tag{41}$$

The ratio $15/\pi^2$ is a pure dimensionless number independent of:

- All SM coupling constants (α, α_s, α_w)
- All particle masses (m_e, m_u, m_H, ...)
- All cosmological parameters (H_0, Ω, Λ, ...)
- The number of degrees of freedom g_* (it cancels exactly)

Its sole origin is the ratio of the Heisenberg minimum energy ($E_{\text{min}} = \hbar/2t_{\text{Pl}}$, giving the numerator 15) to the Stefan-Boltzmann radiation constant ($\pi^2/30$, giving the denominator π^2). This is a *structural* result: it holds for any theory whose pre-emergence state saturates the Heisenberg bound.

15.2 Comparison of derivation inputs

Standard cosmology arrives at the Planck-era energy density via the FRW equation, SM thermodynamics, the measured values of H_0, Ω_b, Ω_r, and approximately 20 additional parameters. TUO arrives at E_{cell} from:

Input	Status
Axiom I (Minkowski background)	Postulate
Axiom II (zero-sum, $\mathrm{Tr}\left[\hat{\rho}\hat{Q}_k\right] = 0$)	Postulate
Heisenberg uncertainty $\Delta E\,\Delta t \geq \hbar/2$	Standard QM
SM effective DOF count $g_* = 106.75$	From experiment

No dynamical equations, no measured cosmological parameters, and no initial conditions beyond the axioms are required.

15.3 Energy scale in context

Table 3 places E_{cell} in the hierarchy of known energy scales. The result derived from two axioms sits at the intersection of quantum mechanics (the Heisenberg bound) and the Planck scale, and exceeds every laboratory energy scale by many orders of magnitude.

Table 3: Energy scale comparisons. $E_{\text{cell}} = 1.044 \times 10^{11}$ J is derived from two axioms alone, with $g_* = 106.75$ as the only experimental input.

Reference scale	Energy (J)	E/E_{cell}
Electron rest mass $m_e c^2$	8.19×10^{-14}	7.8×10^{-25}
Proton rest mass $m_p c^2$	1.50×10^{-10}	1.4×10^{-21}
LHC full beam energy	3.15×10^{-3}	3.0×10^{-14}
1 megaton nuclear yield	4.0×10^{15}	3.8×10^{4}
E_{cell} **(TUO, from axioms)**	$\mathbf{1.044 \times 10^{11}}$	**1**
Planck energy E_{Pl}	1.956×10^{9}	1/53.4
Solar mass-energy $M_\odot c^2$	1.79×10^{47}	1.7×10^{36}

15.4 Significance of the zero-total-energy result

The result $E_{\text{matter}} + E_{\text{grav}} = 0$ (Theorem 5.1) follows from the single identity $GM_{\text{Pl}}^2 = \hbar c$, which is not an empirical coincidence but an exact consequence of how the Planck units are defined. Numerically:

$$\frac{GM_{\text{Pl}}^2}{\hbar c} = \frac{G \cdot (\hbar c/G)}{\hbar c} = 1.000\,000\,000\,000\,000 \tag{42}$$

to 15 significant figures (machine double precision). The energy is not *approximately* zero; it is *algebraically* zero from the definitions of G, $\hbar$, and c.

The physical significance: the universe contains 1.044×10^{11} J of matter energy per Planck cell at emergence, and an equal and opposite -1.044×10^{11} J of gravitational energy. The net is exactly zero. The apparent richness of the universe is a rearrangement of nothing.

15.5 Sensitivity to g_*

Although $E_{\text{cell}}/E_{\text{thermal}} = 15/\pi^2$ is independent of g_*, the absolute energy $E_{\text{cell}} = g_* E_{\text{Pl}}/2$ is linear in g_*. A change of one effective degree of freedom shifts the cell energy by $E_{\text{Pl}}/2 \approx 9.78 \times 10^8$ J, or about 0.94% of the total. For the SM value $g_* = 106.75$,

$$\frac{\partial E_{\text{cell}}}{\partial g_*} = \frac{E_{\text{Pl}}}{2} = 9.780 \times 10^8 \text{ J per effective DOF.} \tag{43}$$

This makes E_{cell} a direct probe of the particle content at emergence: any future measurement of the Planck-era energy density constrains g_* without any other free parameter.

16 Quantitative Predictions

Table 4 assembles the quantitative predictions of TUO that follow strictly from the two axioms plus SM particle content, together with their observational status.

17 Open Problems

Open Question 17.1 (Baryon asymmetry). TUO shows that $B - L = 0$ permits matter-only configurations. It does not derive $\eta = 6.12 \times 10^{-10}$. A full derivation requires QFT baryon-number dynamics at T_{Pl} with Sakharov-type washout factors.

Open Question 17.2 (CMB power spectrum). The saddle-point argument of Section 10 suggests simultaneous emergence, giving $\Delta T/T = 0$ at leading order. Within-cell quantum fluctuations give $\Delta E_{\text{cell}}/E_{\text{cell}} \sim 1/\sqrt{g_*} \approx 9.7\%$. The observed $\Delta T/T \sim 10^{-5}$ requires ~ 4 orders of magnitude of suppression through GR dynamics. The specific calculation required is the two-point stress-energy correlator $\langle T_{\mu\nu}(x) T_{\rho\sigma}(x')\rangle|_{\text{ZS}}$ in a Planck-cell volume, propagated to horizon crossing.

Open Question 17.3 (Energy density mismatch factor $15/\pi^2$). The TUO Heisenberg energy exceeds the FRW thermal energy by $15/\pi^2 \approx 1.52$. Whether this is a physical effect (the pre-emergence state is colder than equilibrium) or an artifact of the junction prescription is undetermined.

Open Question 17.4 (Dark energy). Zero-sum predicts $\Lambda_{\text{eff}} = 0$ at leading order. The residual $\Lambda_{\text{obs}} \approx 10^{-122} E_{\text{Pl}}^4$ is not derived here and is left as an open problem for the companion QGD framework.

Table 4: TUO results from Axioms 1–2 and SM content. Status: ✓✓ = TUO derives what standard cosmology assumes as input; ✓ = quantitatively consistent with observation; ∼ = qualitatively consistent, full calculation needed; ⋆ = new distinguishing prediction; † = open problem.

#	Result	TUO value	Observation	Status
1	Total energy E_{tot}	$= 0$ exactly (Theorem 5.1)	≈ 0 (flat universe)	✓✓
2	Spatial flatness Ω	$= 1$ exactly (Axiom 1)	1.0007 ± 0.0037 [1]	✓✓
3	Equation of state w	$= 1/3$ derived (Sec. 11)	$1/3$ (input to FRW)	✓✓
4	Hubble $H(t_{\text{Pl}})$	$= 1/(2t_{\text{Pl}})$ derived (Sec. 12)	$1/(2t_{\text{Pl}})$ (FRW input)	✓✓
5	Expansion law	$V \propto t^3$, $v < c$ always	$V \propto t^3$ (rad. dom.)	✓✓ + ⋆
6	Energy per cell	$(g_*/2)E_{\text{Pl}} = 1.04 \times 10^{11}$ J	6.87×10^{10} J (thermal)	✓ (factor $15/\pi^2$)
7	Quantum correction	$\Delta V/V = 3(\ell_{\text{Pl}}/ct)^2$	Not yet measured	⋆
8	$B - L = 0$ per generation	SM colour structure (Theorem 8.1)	SM anomaly cancellation	✓
9	Proton stability	$\Delta(B - L) = 0$ channels allowed	$\tau_p > 10^{34}$ yr [?]	⋆ testable
10	Baryon asymmetry	$B = L \neq 0$ permitted	$\eta = 6.12 \times 10^{-10}$	∼ †
11	CMB uniformity	Simultaneous emergence (Sec. 10)	$\Delta T/T \sim 10^{-5}$	∼ †

Open Question 17.5 (SM particle content). TUO uses $g_* = 106.75$ from experiment. Whether SM content follows from the constrained path integral is open.

18 The Hot Big Bang Begins Here

TUO is not a replacement for the Hot Big Bang (HBB) model. It is a complement: TUO describes the pre-emergence phase and delivers the HBB its initial conditions. This section states precisely where TUO ends and the HBB begins, and verifies that the handoff is seamless.

18.1 What the Hot Big Bang requires at $t = t_{\text{Pl}}$

The standard HBB model takes the following as empirical starting conditions at $t = t_{\text{Pl}}$ [7, 8]:

(i) Radiation equation of state: $w = 1/3$

(ii) Hubble rate: $H(t_{\rm Pl}) = 1/(2t_{\rm Pl})$ (radiation domination)

(iii) Spatial flatness: $\Omega = 1$

(iv) Full SM plasma: all $g_* = 106.75$ effective degrees of freedom

(v) Thermal equilibrium at $T \sim T_{\rm Pl}$

None of these are derived within the HBB itself.

18.2 What TUO delivers

Theorem 18.1 (TUO-to-HBB handoff)**.** *The maximum-fluctuation emergence configuration delivers all five HBB initial conditions without additional assumptions.*

Proof. (i) Equation of state. Wavepacket initial condition with $\Delta x = \ell_{\rm Pl}$ gives $\Delta p = M_{\rm Pl}c/2$; relativity gives $v_0 = c$; therefore $w = 1/3$ (Section 11, derived not assumed).

(ii) Hubble rate. $H_{\rm TUO}(t_{\rm Pl}) = 1/(2t_{\rm Pl})$ from the expansion law (Eq. (34)). In radiation-dominated FRW, $H = 1/(2t)$ exactly. Both are equal at $t = t_{\rm Pl}$.

(iii) Spatial flatness. Axiom 1 gives $k = 0$ exactly.

(iv) Full SM plasma. $E_{\rm cell} = g_* E_{\rm Pl}/2 \approx 1.04 \times 10^{11}$ J exceeds the pair-production threshold $2m_f c^2$ for every SM particle. The heaviest (top quark, $m_t \approx 173\,\text{GeV}$) has threshold 5.46×10^{-8} J; the ratio is $E_{\rm cell}/(2m_t c^2) \approx 1.9 \times 10^{18}$. All particle species are produced instantaneously.

(v) Thermal equilibrium. All g_* modes emerge simultaneously with the same Heisenberg energy $E_{\rm Pl}/2$. This is, by definition, a thermal distribution at $T_{\rm TUO} = (15/\pi^2)^{1/4} T_{\rm Pl} \approx 1.11\, T_{\rm Pl}$. No further thermalisation is required. □

18.3 The junction

TUO ($t < t_{\rm Pl}$)	**at** $t = t_{\rm Pl}$	**Hot Big Bang** ($t > t_{\rm Pl}$)
Zero-sum pre-emergence	$\longleftrightarrow$	Radiation domination
Wavepacket $\sigma(t)$, $v < c$		$a(t) \propto t^{1/2}$
$w = 1/3$ (derived)	$w = 1/3$	$w = 1/3$ (assumed)
$H = 1/(2t_{\rm Pl})$ (derived)	H continuous	$H = 1/(2t)$
$k = 0$ (Axiom I)	$\Omega = 1$	$\Omega = 1$ (assumed)
$g_* = 106.75$ (all SM)	plasma established	HBB thermodynamics

Remark 18.2. The energy density mismatch by $15/\pi^2 \approx 1.52$ (Section 15) does not break the continuity of H, w, or k, since H is derived from the expansion law independently

of temperature. As the initial maximum-fluctuation state thermalises, the effective temperature relaxes to the value consistent with the Friedmann equation. The resolution of this factor is Open Question 17.3.

19 Summary

The Theory of Universal Origins is a pre-Big-Bang framework based on two axioms applied to the infinite-dimensional Fock space of Standard Model fields. Its central object is the infinite zero matrix $\mathbf{0}_\infty$; its central equation is $\mathbf{Q}[\hat{\rho}] = \mathbf{0}_\infty$.
Derived results (from axioms alone):

1. $w = 1/3$ — the radiation equation of state
2. $H(t_{\rm Pl}) = 1/(2t_{\rm Pl})$ — the Planck-era Hubble rate
3. $\Omega = 1$ — exact spatial flatness
4. $V(t) \propto t^3$, $v < c$ always — the expansion law
5. $E_{\rm cell} = (g_*/2)E_{\rm Pl}$, $E_{\rm grav} = -E_{\rm cell}$ — zero total energy
6. No-annihilation theorem — matter-only stability
7. $B - L = 0$, $Q = 0$ per SM generation — anomaly identity
8. $T_{\rm TUO} = (15/\pi^2)^{1/4} T_{\rm Pl}$, independent of g_* — universal temperature
9. All five HBB initial conditions — seamless handoff

Open problems for future work: baryon asymmetry η, CMB power spectrum amplitude, the $15/\pi^2$ energy-density factor, dark energy, SM particle content from axioms, derivation of $N_{\rm gen} = 3$.

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References

[1] Planck Collaboration (N. Aghanim et al.), "Planck 2018 results VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020). doi:10.1051/0004-6361/201833910.

[2] Planck Collaboration (Y. Akrami et al.), "Planck 2018 results X. Constraints on inflation," *Astron. Astrophys.* **641**, A10 (2020). doi:10.1051/0004-6361/201833887.

[3] R. L. Workman et al. (Particle Data Group), "Review of Particle Physics," *Prog. Theor. Exp. Phys.* **2022**, 083C01 (2022). doi:10.1093/ptep/ptac097.

[4] A. H. Guth, "Inflationary universe: A possible solution to the horizon and flatness problems," *Phys. Rev. D* **23**, 347 (1981). doi:10.1103/PhysRevD.23.347.

[5] A. D. Linde, "A new inflationary universe scenario," *Phys. Lett. B* **108**, 389 (1982). doi:10.1016/0370-2693(82)91219-9.

[6] Super-Kamiokande Collaboration (K. Abe et al.), "Search for proton decay via $p \to e^+\pi^0$," *Phys. Rev. D* **95**, 012004 (2017). doi:10.1103/PhysRevD.95.012004.

[7] S. Weinberg, *Cosmology* (Oxford University Press, Oxford, 2008).

[8] E. W. Kolb and M. S. Turner, *The Early Universe* (Addison-Wesley, Reading MA, 1990).

[9] D. Lovelock, "The Einstein tensor and its generalizations," *J. Math. Phys.* **12**, 498 (1971). doi:10.1063/1.1665613.

[10] E. P. Tryon, "Is the Universe a Vacuum Fluctuation?" *Nature* **246**, 396 (1973). doi:10.1038/246396a0.

[11] E. Noether, "Invariante Variationsprobleme," *Nachr. Ges. Wiss. Göttingen* 235 (1918). [arXiv:physics/0503066]

[12] R. Matshaba, "Universal Origins: The Zero-Sum Constraint," preprint, University of South Africa (2025).